GRANGER CAUSALITY AND NETWORK THEORY

Reading Time	60 minutes
Prior Knowledge	OLS, Covariance Matrix, VAR model, Basic Network Theory, Basic Python
Keywords	Association, Causality, Granger Causality

1. Introduction

In the last lesson, we talked a lot about one variable influences another variable in a network many times. In this lesson, we are going to go one step further and discuss various types of relationships among variables. We will first introduce the concept of association among variables and its characteristics. Next, we will give a short explanation about causality. Then we will introduce Granger causality and explain its assumptions and limitations. We will also provide some Granger causality application examples. Finally, we will show how to combine Granger causality with network to analyze relationship among time series.

2. Association Versus Causality

2.1. Understand Association

Association or **correlation** is used to describe the statistical relationship or pattern of two time series(Nicolae et al, 2022). The correlation coefficient of two time series tells us how strong and what direction of the association of the two time series.

Some important properties of the association are listed as follows:

1. The relationship is symmetric, which means if X is associated with Y, Y is also associated with X
2. Association does not indicate if one variable influences or has direct connection to the other variable. We also call it non-directional
3. Association can exist even two variables do not have direct causal connection
4. Association can be measured using correlation coefficients

If the correlation coefficient shows that two time series are highly associated (correlated), our next question is usually if one time series could cause the other time series. There can be two possible causal scenarios:

1. Direct Causation: change in variable X causes change in variable Y (or vice-versa)
2. Common response: variable X and variable Y are influenced by variable Z

We can use the network graph we learned in the last lesson to demonstrate the two scenarios.

Figure 1. Network Graph for Causal Types of Association
M7_L2_network_graph

From figure 1, we can see Direct Causation situation on the left is the most straight forward causal relationship. Common Response situation on the right is the same as the complicated case we had with a confounding variable C in the last lesson. When a confounding variable is not included in a model, the model result will create a spurious association. Identifying and including a confounding variable is best practice to remove the spurious association between A and B.

2.2. Understand Causality

Causality means that one variable directly influences or makes changes in another (Hansen, 2014). Causal Inference is the method to analyze actual cause-and-effect relationship among variables.
A variable x_1 is considered to have a causal influence on the outcome variable y if changing x_1 leads to a change in y, assuming all other variables remain fixed. The complete model for the outcome can be expressed as:

$$y = h(x_1, x_2, u)$$

where x_1 and x_2 are observable inputs,
u is an unmeasured random component,
h describes the functional relationship among these elements.

The causal effect of x_1 is defined as the rate of change in y with respect to x_1, while keeping x_2 and u constant:

$$C(x_1, x_2, u) = \nabla_1 h(x_1, x_2, u)$$

where ∇_1 is the partial derivative with respect to x_1.
Here are the key properties of causality:

1. Causality represents asymmetric relationship, which means even X causes Y, it doesn't mean Y causes X
2. Causality shows which variable influences the other. It is also called directional
3. Causality is sequential, which means cause precedes effect

4. There is a mechanism to make influence happen

3. Granger Causality

3.1. Introduction

In the last section, we discussed association and causality. In this section, we are going to introduce another related method to analyze the relationship of two time series - Granger Causality. **Granger Causality** is a framework introduced by Granger(1969) to access if the past values of one time series can predict the other time series. For example, Granger causality is used to check if the historical values of x_t is predictive of y_t. If the answer is yes, we say x Granger causes y. Granger causality leverages the feature that there is an ordering of values in a time series to study cause-and-effect relationship for time series.
One thing we need to be careful is even x Granger causes y does not mean x directly causes y. Granger causality is different from the causality we discussed in the last section. Granger causality focuses more on using the history of one variable to predict another variable and not actual cause-and-effect analysis.
Researchers have been using Granger causality to find out interactions among variables in finance and macroeconomics. For example, Billio et al (2012) used Granger causality to understand the connectivity and systemic risk of financial institutions. However, there are limitations about Granger causality to provide causal information among time series. We will discuss these limitations in the later part of this section. First, let's give a formal description of Granger Causality.

3.2. Concept of Granger Causality

We have two **stationary** time series x_t and y_t. With these two time series, we can define the following two information sets:

$$\mathcal{H}_{1,t} = (y_t, y_{t-1}, y_{t-2}, \dots)$$

$$\mathcal{H}_{2,t} = (y_t, x_t, y_{t-1}, x_{t-1}, y_{t-2}, x_{t-2}, \dots)$$

where information set $\mathcal{H}_{1,t}$ includes information from y_t only and information set $\mathcal{H}_{2,t}$ includes information from both y_t and x_t.

We say x 'Granger causes' y if

$$var[y_t - \mathcal{P}(y_t|\mathcal{H}_{2,t-1})] < var[y_t - \mathcal{P}(y_t|\mathcal{H}_{1,t-1})] \qquad (1)$$

where $\mathcal{P}(y_t|\mathcal{H}_{2,t-1})$ is the optimal prediction of y_t, given the information set $\mathcal{H}_{2,t-1}$ and $\mathcal{P}(y_t|\mathcal{H}_{1,t-1})$ is the optimal prediction of y_t, given the information set $\mathcal{H}_{1,t-1}$

Equation (1) says when the variance of prediction errors of y_t given information set $\mathcal{H}_{2,t-1}$ is smaller than the variance of prediction errors of y_t given information set $\mathcal{H}_{1,t-1}$, the time series x Granger causes y. It means the addition of past values of x_t

in the model improves the prediction of y_t and reduces the variance of prediction errors.
Based on the above framework, how can we implement the actual Granger causality test? We can start by writing down the following two OLS regression models:

(reduced model)
$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} \ldots + \alpha_p y_{t-p} + \epsilon_{t(\mathcal{H}_1)}$

(full model)
$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} \ldots + \alpha_p y_{t-p} + \beta_1 x_{t-1} + \beta_2 x_{t-2} \ldots + \beta_p x_{t-p} + \epsilon_{t(\mathcal{H}_2)}$

where α_i are coefficients of y_i and β_j are coefficients of x_j while $i, j \in [1, 2, \ldots, p]$
$\epsilon_{t(\mathcal{H}_1)}$ is the white noise for the reduced model $\epsilon_{t(\mathcal{H}_2)}$ is the white noise for the full model
Equation (2) represents the model that only uses the past values of y to predict y_t while equation (3) represents the model that uses not only the past values of y but also the past values of x to predict y_t . Equation (2) uses information set $\mathcal{H}_1$ hence this model is a reduced model only using partial information. Equation (3) uses information set $\mathcal{H}_2$ therefore this model is a full model fusing full information. To check if x Granger causal to (or Granger causes) y , we can write down the following hypothesis testing:

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_p = 0$$

$$H1 : \text{at least one of the } \beta s \neq 0$$

To test the above hypothesis, we will use F-test. Our test statistic is as follows:

$$F = \frac{(RSS_{reduced} - RSS_{full})/(r - s)}{(RSS_{full})/(T - r)}$$

where RSS is residual sum of squares of the regression model
r is the number of parameters in the full model
s is the number of parameters in the reduced model
T is the number of samples

If the p value of the F-test is less than 5% given the significance level is set at 5%, we can reject the null hypothesis(H_0) and say x Granger causes y. We will then switch the role of x and y to repeat the above said procedure to check if y Granger causes x.

3.3. Vector Autoregression Model(VAR) and Granger Causality

In the last section, we described the detailed process of how to test if x Granger causes y. And then use the same process to test if y Granger causes x. Many times,

we can present the last two Granger causality tests in a bivariate VAR(p) framework. Let's first write down a VAR model for x and y with lag of p.

$$x_t = \phi_0 + \phi_1 x_{t-1} + \phi_2 x_{t-2} \ldots + \phi_p x_{t-p} + \theta_1 y_{t-1} + \theta_2 y_{t-2} \ldots + \theta_p y_{t-p} + \varepsilon_t \quad (4$$

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} \ldots + \alpha_p y_{t-p} + \beta_1 x_{t-1} + \beta_2 x_{t-2} \ldots + \beta_p x_{t-p} + \epsilon_t \quad ($$

Then we estimate this VAR(p) model. Once we have the coefficient estimates of the model, we can conduct Granger causality test. For example, if we want to test whether x Granger causes y, it is equal to test $H_0 : \beta_1 = \beta_2 = \ldots = \beta_p$. If we want to test whether y Granger causes x, it is equal to test $H_0 : \theta_1 = \theta_2 = \ldots = \theta_p = 0$. Another reason we will use VAR(p) framework for Granger causality test is the selection of the number of lags p. That leads to our next topic: the assumptions of Granger causality.

3.4. Assumptions and Limitations of Granger Causality

Now you have seen how to set up a test for Granger causality for two time series. You might also have seen a few assumptions you used during the description of the process. Let's summarize the key assumptions here.

Stationarity
The two time series need to be stationary before applying Granger causality test. We can use augmented Dickey-Fuller test to check for stationarity for the time series. If any time series is not stationary, we can apply the methods we learned from previous lessons to transform the time series.
Linearity
Granger causality test assumes the time series are in a linear relationship.
Continuous-Valued Time Series
The values of the time series for Granger causality test need to be continuous.
No Confounding Variables
All relevant variables are observed and included in the analysis and there are no unmeasurable variables.
Lag is known
The lag for the dependent past observations is known and uniform across all variables.

The linear dependency of the time series assumption can be restrictive as many time series data have non-linear relationship. The no confounding variables assumption is unrealistic especially for the bivariate model we having been discussing. We talked about the problem of non including confounding variables in the model in the last lesson. We will demonstrate this issue when we show the Granger causality and network combination example. As for known lag assumption, we can use the lag

order selection function in the VAR model to decide the optimal lag for Granger causality test.

3.5. An Example for Granger Causality Test in Bivariate VAR model

In this section, let's try to run Granger causality for just two time series. We will pick returns of JPMorgan stock and Technology ETF for this example. As discussed in the last section, we need to check for stationarity of these two time series before conducting Granger causality test. We will use augmented Dickey Fuller test to check for stationarity.

In [1]:
```
# pip install networkx seaborn --upgrade
```

In [2]:
```
import numpy as np
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import yfinance as yf
import warnings
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.api import VAR

warnings.filterwarnings("ignore")
```

In [3]:
```
# Gather stock data and factor data for 7 years

# List of assets (stocks)
assets = ['AAPL',   # Apple
          'WMT',    # Walmart
          'AMZN',   # Amazon
          'GOOGL',  # Google
          'JPM',    # JP Morgan
          'WBD',    # Warner Brothers
          'GE']     # General Electric Co.

# List of factors
factors = [
    '^GSPC',   # S&P 500 (market factor)
    'XLF',     # Financial Select Sector SPDR Fund
    'XLK',     # Technology Select Sector SPDR Fund
    'SHY',     # Short-term Treasury ETF (interest rate factor)
    'DX-Y.NYB' # US Dollar index
]

# Combine assets and factors
tickers = assets + factors

# Download data
data_0 = pd.read_csv('M7L2_stock_prices.csv')

data = data_0.drop('Date', axis=1)

# Calculate returns
returns = data.pct_change().dropna()
```

```
# Add meaningful labels
returns.columns = [
    'Apple', 'Amazon', 'USD', 'GE', 'Google', 'JPMorgan', 'Treasury', 'Wa
]
```

In [4]:
```
# Standardize data
standardized_returns = (returns - returns.mean()) / returns.std()
```

In [5]:
```
# Perform the ADF test for JPMorgan stock return
jp_result = adfuller(returns['JPMorgan'])

print('p-value of JPMorgan: %f' % jp_result[1])
```

```
p-value of JPMorgan: 0.000000
```

In [6]:
```
# Perform the ADF test for Technology ETF
tech_result = adfuller(returns['Technology'])

print('p-value of Technology: %f' % tech_result[1])
```

```
p-value of Technology: 0.000000
```

From the results of the above augmented Dickey Fuller tests, both time series are stationary. Next we will use select lag function from VAR model to find the optimal lag order for Granger causality test.

In [7]:
```
# Use optimal lag order to run Granger Causality test
from statsmodels.tsa.api import VAR

# Create a dataset to run the VAR model
gc_data = returns[["JPMorgan","Technology"]]

# Use select_order to find the optimal lag order
model = VAR(gc_data)
lag_order_results = model.select_order(maxlags=10)
lag_order = lag_order_results.aic # You can also use bic, fpe or hqic
print('Optimal Lag Order:', lag_order)
```

```
Optimal Lag Order: 9
```

The best lag order is 9 based on the above code result. With that information, now we can run the VAR model with the best lag order and run Granger causality test.

In [8]:
```
# Fit the VAR model
model_fitted = model.fit(lag_order)
```

With the fitted model, we can now to test Granger causality for time series 'Technology' and time series 'JPMorgan'. We will test both directions.

In [9]:
```
# Perform Granger causality test on JPMorgan Granger causes Technology; t
test_result = model_fitted.test_causality(caused='Technology', causing='J
print(test_result.summary())
```

```
Granger causality F-test. H_0: JPMorgan does not Granger-cause Technology.
Conclusion: reject H_0 at 5% significance level.
==========================================================
Test statistic Critical value p-value          df
----------------------------------------------------------
         2.353          1.883   0.012 (9, np.int64(3462))
----------------------------------------------------------
```

In [10]:
```
# Perform Granger causality test on Technology Granger causes JPMorgan; t
test_result = model_fitted.test_causality(caused='JPMorgan', causing='Tec
print(test_result.summary())
```

```
Granger causality F-test. H_0: Technology does not Granger-cause JPMorgan.
Conclusion: reject H_0 at 5% significance level.
==========================================================
Test statistic Critical value p-value          df
----------------------------------------------------------
         3.105          1.883   0.001 (9, np.int64(3462))
----------------------------------------------------------
```

From the above Granger causality test results, we can see that JPMorgan and Technology ETF Granger cause each other. These two time series have dynamic interactions with each other.

4. Granger Causality and Network Theory

In this section, we are going to combine Granger causality with network theory to study interconnection of multiple time series. We will have stock and factor returns as nodes. Their link will be determined by Granger causality. If asset X Granger causes asset Y, there will be an arrow pointing to asset Y from asset X. Then we can draw the network graph to understand the interconnections among assets and factors based on Granger causality.
Let's begin by checking the stationarity for all time series in the dataset.

In [11]:
```
# Check for Stationarity for all time series
results = {'Column': [], 'P-Value': [], 'Stationary (5% level)': []}

# Loop through each column
for column in returns.columns:
    adf_result = adfuller(returns[column])

    # Extract p-value (second value in the result tuple)
    p_value = adf_result[1]

    # Add results to dictionary
    results['Column'].append(column)
    results['P-Value'].append(p_value)
    results['Stationary (5% level)'].append('Yes' if p_value < 0.05 else

results_df = pd.DataFrame(results)
print(results_df)
```

```
        Column       P-Value Stationary (5% level)
0        Apple  9.268138e-25                   Yes
1       Amazon  0.000000e+00                   Yes
2          USD  2.046682e-30                   Yes
3           GE  2.764974e-23                   Yes
4       Google  3.329989e-26                   Yes
5     JPMorgan  1.046926e-22                   Yes
6     Treasury  2.080702e-14                   Yes
7   Warner-Bro  0.000000e+00                   Yes
8      Walmart  1.088815e-26                   Yes
9    Financial  3.848073e-24                   Yes
10  Technology  1.931759e-25                   Yes
11       SP500  3.877946e-24                   Yes
```

From the above result, we can see all asset and factor returns are stationary. Now we can move to the next step. We will follow the Granger causality test process for bivariate VAR model example in the last section and loop through all the pairs in the dataset. If the Granger causality test is statistically significant for a pair of time series, the entry of the adjacency matrix of this pair will be 1. Otherwise ,the entry will be 0. We also apply Bonferroni method to adjust P value for statistical level. Finally, we will use the filled adjacency matrix to draw the network graph.

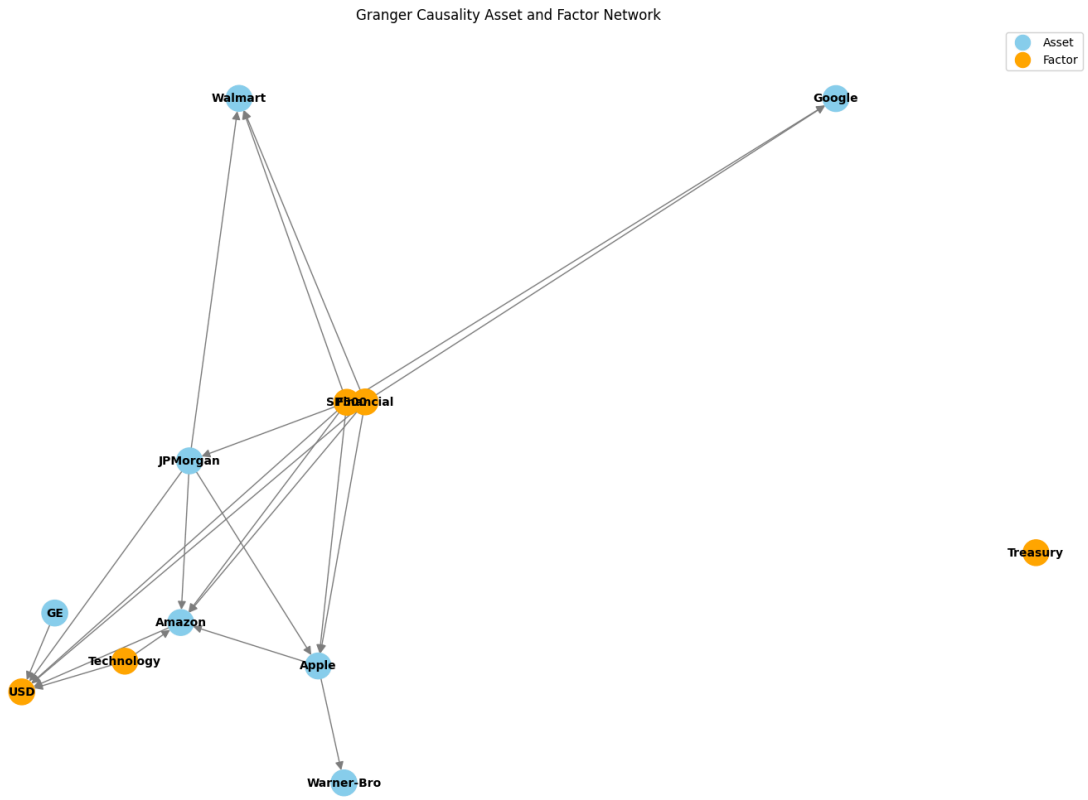
In [12]:

```
# Run a pairwise VAR model and then Granger Causality Asset and Factor Ne
n_var = len(returns.columns)
adj_matrix = np.zeros((n_var, n_var))

# Significance threshold (adjusted based on Bonferroni Method)
p_threshold = 0.05/(n_var*(n_var-1))

# Loop through all pairs of assets
for i, return_i in enumerate(returns):
    for j, return_j in enumerate(returns):
        if i != j:  # Skip self-relationships
            var_data = returns[[return_i, return_j]]
            # Use select_order to find the optimal lag order
            model = VAR(var_data)
            lag_order_results = model.select_order(maxlags=10)
            lag_order = lag_order_results.aic # You can also use bic, fpe
            if lag_order > 0:
              final_order = lag_order
            else:
              final_order = 1
            # Fit the VAR model
            model_fitted = model.fit(final_order)
            # Run conditional Granger Causality
            test_result = model_fitted.test_causality(caused= return_j, c

            # Get p-value for the Granger causality test
            p_value = test_result.pvalue

            # If significant, add connection to adjacency matrix
            adj_matrix[i, j] = 1 if p_value < p_threshold else 0

# Create directed graph from adjacency matrix
G = nx.from_numpy_array(adj_matrix, create_using=nx.DiGraph)

# Relabel nodes with asset names
```

```
mapping = {i: name for i, name in enumerate(returns)}
G = nx.relabel_nodes(G, mapping)
```

In [13]:

```
# Create and visualize the network

# Assign types to nodes
node_types = {
    'Apple':'Asset',
    'Amazon':'Asset',
    'USD':'Factor',
    'GE':'Asset',
    'Google':'Asset',
    'JPMorgan':'Asset',
    'Treasury':'Factor',
    'Warner-Bro':'Asset',
    'Walmart':'Asset',
    'Financial':'Factor',
    'Technology':'Factor',
    'SP500':'Factor',
}

nx.set_node_attributes(G, node_types, 'type')

# Draw the network graph
plt.figure(figsize=(14, 10))

# Use different colors for assets and factors
node_colors = ['skyblue' if G.nodes[name]['type'] == 'Asset' else 'orange

# Create layout
pos = nx.spring_layout(G, seed=42)

# Draw the network
nx.draw(G, pos,
        with_labels=True,
        node_size=500,
        node_color=node_colors,
        font_size=10,
        font_weight='bold',
        edge_color='gray',
        arrows=True,
        arrowsize=15)

# Add a legend
asset_patch = plt.Line2D([0], [0], marker='o', color='w', markerfacecolor
                         markersize=15, label='Asset')
factor_patch = plt.Line2D([0], [0], marker='o', color='w', markerfacecolo
                          markersize=15, label='Factor')
plt.legend(handles=[asset_patch, factor_patch], loc='upper right')

plt.title('Granger Causality Asset and Factor Network')
plt.show()
```



The above network graph provides some interesting Granger causality test results. From the graph, we can see SP500 and Financial ETFs Granger cause all the stock returns except GE and Warner Brothers. One interesting observation is that several stocks also Granger cause USD ETF. For example, not only SP500 Granger causes USD ETF, Amazon also Granger causes USD ETF. There might be a spurious relationship going on here. As we discuss before Granger causality test in bivariate VAR model setting can have no confounding variable issue. In the above graph, we see SP500 Granger causes USD ETF. SP500 Granger also Granger causes Amazon. Then Amazon also Granger causes USD ETF. SP500 might be the confounding variable when we conduct Granger causality test between Amazon and USD ETF. Let's check on that by running the following 3 variable VAR model. As before, we will find the optimal lag order and fit the model. Once the model is fitted, we can conduct Granger causality test.

In [14]:
```
# Create a dataset to run the VAR model
gc_data = returns[["Amazon","SP500","USD"]]

# Use select_order to find the optimal lag order
model = VAR(gc_data)
lag_order_results = model.select_order(maxlags=10)
lag_order = lag_order_results.aic # You can also use bic, fpe or hqic
# Fit the VAR model
model_fitted = model.fit(lag_order)
```

In [15]:
```
# Perform Granger causality test on JPMorgan Granger causes Technology; t
test_result = model_fitted.test_causality(caused='USD', causing='Amazon',
print(test_result.summary())
```

```
Granger causality F-test. H_0: Amazon does not Granger-cause USD. Conclusi
on: fail to reject H_0 at 5% significance level.
============================================================
Test statistic Critical value p-value         df
------------------------------------------------------------
         1.673          1.882   0.090 (9, np.int64(5166))
------------------------------------------------------------
```

From the above Granger causality test result, we see that P value is 0.09 and we cannot reject H_0 and the result is Amazon does not Granger cause USD ETF. This is the example how omitting a confounding variable can change the result of Granger causality test. In the recent research advance, searchers have proposed several modifications to handle no confounding variable issue. We will not cover them in this lesson. For those who are interested, there is plenty information online.

5. Conclusion

In this lesson, we learned the differences between association and causality. Then we introduced Granger causality and went through its theory. We also discussed the assumptions and limitations of Granger causality. Finally we combined Granger causality and network theory to produce a Granger causality based network. This complex network can help us to better understand the Granger causal relationship among assets and factors.

References

1. Nicolae et al, "Part I: Exploring Data, 10. Data Collection, Causality and Association", Introduction to Data Science, University of Chicago, 2022, https://ds1.datascience.uchicago.edu/10/data-collection.html
2. Hansen "Conditional Expectation and Projection: Causal Effect ", Econometrics, University of Wisconsin, 2014, pp. 43
3. Billio, M., Getmansky, M., Lo, A. W. and L. Pelizzon. (2012) Econometric measures of connectedness and systemic risk in the finance and insurance sectors. Journal of Financial Economics, 104(3), pp. 535–559.

Copyright 2025 WorldQuant University. This content is licensed solely for personal use. Redistribution or publication of this material is strictly prohibited.

In [ ]: